

USMAN, ABDULLAHI IDRIS

| [Google Scholar](#) | Phone No: +905359379120 | Sakarya, Türkiye | [LinkedIn](#) | [Email](#)

Research Interests

Non-linear solid and structural mechanics; finite element modelling under compressive or impact loading conditions; crashworthiness and energy-absorbing potential of lattice and composite structures; additive manufacturing of architected materials; experimental–numerical correlations and mechanical characterization.

Education

M.Sc. - Mechanical Design and Manufacturing | CGPA: 3.5/4.0 (Top 5%) *Sep 2025 - Jul 2027 (Expected)*
Sakarya University, Sakarya, Türkiye - [Link](#) Distinction

- **Thesis:** *Investigation of the Impact Resistance of Vacuum-Formed and PU Resin Foam-Reinforced 3D-Printed Chain-Fabrics.*
- **Supervisor:** Asst. Prof. Dr. Muhammet Muaz Yalçın

Exchange Student - Mechanical Engineering (Erasmus+) *Feb 2027 - May 2027 (Expected)*
• VŠB – Technical University of Ostrava, Ostrava, Czech Republic (*Erasmus+ Scholar*)

Turkish Language Preparation Programme | CEFR Level: C1 *Oct 2024 - Aug 2025*
Sakarya University Turkish Language Teaching Application and Research Center (TÖMER) - [Link](#)

- *Completed intensive Turkish-language training to prepare for graduate study in Türkiye; attained C1 level.*

B.Eng. - Mechanical Engineering | CGPA: 4.5/5.0 (Scholaro-converted: 3.7/4.0) (Top 1%) *Mar 2017 - Sept 2023*
Abubakar Tafawa Balewa University (ATBU), Bauchi, Nigeria - [Link](#) First Class Honors

- **Thesis:** *Development and Characterization of Doum Palm Shell–Recycled LDPE Composite for Crash Helmet Applications*

Publications and Disseminations

Journal Articles

1. **Idris, U.A.**, Yalçın, M.M. “Functionally Graded Polymer Lattice Structures Enabled by Multi-Material Fused Deposition Modeling: Current Progress, Challenges, and Future Perspectives.” *Under review at MDPI Polymers*, 2026.
2. **Idris, U.A.**, Babuje, A.U., Lamidi, M., Aman, N.D., Amon, S.G. “Mechanical Performance of Doum Palm Shell–Recycled LDPE Hybrid Composites for Safety Helmet Applications.” *Submitted*, 2025.

Research and Teaching Experience

M.Sc. Researcher - Structural Mechanics & Additive Manufacturing *Sept 2025 - Present*
Sakarya University - MMY Structural Mechanics & AM Research Lab Sakarya, Türkiye

- Develop finite-element models in LS-DYNA to predict quasi-static and/or dynamic deformation, force–displacement response, absorbed energy, and failure-critical regions.
- Investigate the impact response of vacuum-formed and PU-resin-foam-reinforced 3D-printed chain-fabric structures for lightweight energy-absorption applications.
- Fabricate polymer specimens by FDM and conduct impact tests; validate numerical predictions against experimental force–displacement and deformation data.
- Designed convex and concave lattice structure topologies as an extension to conventional lattice structures for energy-absorption applications.
- Reviewed and synthesized literature on functionally graded lattice structures, compiled into a review manuscript.

Graduate Assistant - Department of Mechanical and Production Engineering *Nov 2023 - Oct 2024*
Kaduna Polytechnic, Kaduna under National Youth Service Corps (NYSC) Kaduna, Nigeria

- Assisted in conducting laboratory sessions and workshop training for students.
- Provided support to lecturers by helping with tasks such as course preparation, grading, and classroom activities.

Undergraduate Researcher - Sustainable Composite Materials *Nov 2022 - Aug 2023*
Abubakar Tafawa Balewa University (ATBU) Bauchi, Nigeria

- Led a 9-month experimental programme developing natural fiber-recycled polymer composites, testing 5 formulations and achieving 66.67 kJ/m² peak impact strength.

Tutor — Department of Mechanical and Production Engineering *Nov 2022 - Aug 2023*
Abubakar Tafawa Balewa University (ATBU) Bauchi, Nigeria

- Taught Heat Transfer, Strength of Materials III, Machine Design II, and Thermodynamics to course mates.

Selected Engineering Projects

TEKNOFEST Unmanned Surface Vehicle - Design & Additive Manufacturing Lead *Sept 2024 - Present*
ADASTRA - Sakarya University Sakarya, Türkiye

- Led mechanical CAD design and FDM prototyping of 2 hull components, electronics enclosure, and wings for USV.

- Co-developed a successful proposal that secured £50,000 in Teknofest project funding for prototype fabrication, materials, and integration.

TEKNOFEST Roboligue - Mechanical Design & Prototyping Team Member

Sept 2024 - Present

ADASTRA - Sakarya University

Sakarya, Türkiye

- Designed and fabricated 3D-printed components for a robotic car and arm, translating functional requirements into manufacturable CAD models.
- Supported integration and testing of the prototype for the national Teknofest competition submission.

Technical Skills

FEA / Simulation: ANSYS Mechanical, LS-DYNA (Explicit Dynamics for Impact & Crashworthiness Simulation)

CAD / Design: SolidWorks (CSWA & CSWP), NX Siemens

Additive Manufacturing: FDM process parameters, TPMS/BCC/FCC lattice topology design, architected material optimization

Programming: MATLAB, Python

Research Tools: LaTeX, Zotero, MS Office Suite, technical report writing, systematic literature review

Languages: English (C1), Turkish (C1), Hausa (Native), Arabic (A2)

Selected Honors, Awards and Scholarships

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| • Türkiye Bursları (YTB) Scholarship - Link | 2024 |
| • Boston University College of Engineering Graduate Scholarship - Link | 2024 |
| • EducationUSA Opportunity Funds Program - Link | 2024 |
| • Vice-Chancellor's List - Link | 2023 |
| • Best Graduating Student - Prof. Adisa Bello's Award for Best Graduating Student - Link | 2023 |
| • Dean's List (2017–2022, 4 consecutive years) - Link | 2018 |
| • Royal Academy of Engineering - Forging Africa's Future Mechanical Engineers (FAFME) Programme Finalist (Top team from 43 across Nigeria - Link) | 2020 |
| • Chevron JV Scholarship - Link | 2019 |
| • Agbami Medical & Engineering Professional Scholarship - Link | 2018 |

Professional and Industrial Experience

Summer Mechanical Engineering Intern

Aug 2025 - Sep 2025

Sentiürkler Makina

Sakarya, Türkiye

- Operated CNC machines with 10–15 daily tool changes, inspected parts to ± 0.02 mm using coordinate measurement, and updated 3+ SolidWorks drawings, improving shop floor documentation and traceability.

CAD & Simulation Analyst

Jul 2024 - Jan 2025

Etona Engineering Services

Remote

- Conducted static and dynamic FEA on 5+ engineering structures (ANSYS Mechanical), identifying failure-critical zones; performed modal and harmonic analysis on 2+ rotating systems.

Mechanical Engineering Intern

Jun 2022 - Nov 2022

Defense Industries Corporation of Nigeria

Kaduna, Nigeria

- Characterized 12+ metallic and ammunition samples using Spectro Maxx OES, Rockwell/Vickers hardness testing, and optical microscopy, contributing to a 15% reduction in test turnaround time.

Mechanical Engineering Intern

Feb 2020 - Mar 2020

Peugeot Automobile Nigeria (PAN)

Kaduna, Nigeria

- Supported the assembly of a 19-seat bus engine and Peugeot 301 axles, and MIG-welded stretcher rails for ambulance vehicles, gaining hands-on experience in automotive assembly and fabrication.

Leadership and Volunteering Service

Mentor - Northern Elites Scholars Forum - [Link](#)

June 2021 - Present

Founder & Technical Lead - ATBU CAD Club - [Link](#)

Feb 2021 - Sep 2023

President - NIMechE ATBU Student Branch - [Link](#)

Nov 2019 - Feb 2021

Selected Professional Development and Certifications

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|---|------|
| • Additive Manufacturing Specialization - Arizona State University - Link | 2024 |
| • Digital Manufacturing & Design Technology Specialization – University of Buffalo - Link | 2024 |
| • MATLAB Programming for Engineers Specialization - Vanderbilt University - Link | 2024 |
| • Python for Everybody - University of Michigan - Link | 2024 |

Conference, Seminars and Workshops

- Harnessing the Power of Industry 4.0: Transforming Manufacturing with Smart Technologies — EIT Webinar (2024).
- Article Publication in Reputable Journals Workshop — Maryam Abacha American University of Nigeria (2024).

- Nigerian Institution of Mechanical Engineers (NIMechE) 34th Annual International Conference (2021).

Professional Membership and Affiliations

- American Society of Mechanical Engineers (ASME)
- Nigerian Society of Engineers (NSE)
- Nigerian Institution of Mechanical Engineers (NIMechE)
- International Association of Engineers (IAENG)
- Institution of Mechanical Engineers (IMechE)